UNIT 2 · TOPIC 7(A)

Design of Fertilizer Applicators

Syllabus topic: Design of fertilizer applicators and inter-cultivation equipment

Metering difficulty: fertilizers are hygroscopic, corrosive, variable in density/flowability, and prone to caking/bridging — the metering must tolerate all of this (Kepner Ch. 13).

Design requirements (Kepner)

- Uniform rate, insensitive to hopper level and field slope/roughness.
- Wide rate range in small increments, with a reference scale.
- Not damaged by — or damaging to — the fertilizer.
- Corrosion-resistant.
- Easy cleaning.
- Positive shut-off.
- Agitation to prevent bridging.

Metering devices

Feed	How it works / rate control
Star-wheel feed	Rotating star wheels drag fertilizer under an adjustable gate; used extensively on grain-drill fertilizing units, row-crop attachments and full-width broadcasters. Rate = f(wheel size — interchangeable, wheel speed, gate opening).
Auger (screw) feed	Rate ∝ screw speed × pitch volume; uniform for granular products; flighting wear/corrosion is the design issue.
Belt/chain-and-slat feed with gate (bulk spreaders)	Rate = belt speed × gate area × bulk density.

Spinning-disc broadcaster design

- A particle leaves the disc with a velocity set by disc radius and rpm (**$v = \omega r$** at the tip).
- The single-pass pattern is triangular/oval, so the effective swath is the distance at which adjacent-pass overlap gives a uniform total; overlap **25–35%**.
- Vane angle and drop-point position control pattern symmetry.
- Granule size segregation (fines fall short) is the main source of non-uniformity.

Band placement

Place the band **5 cm × 5 cm** from the seed (below-side). This avoids salt injury while giving early access. Use a separate fertilizer tube and boot on the same opener, or a dedicated opener.

Materials

Stainless steel, epoxy-coated steel and engineering plastics (PP/HDPE) for hoppers and metering parts.

Practice MCQs

Q126. Fertilizer metering is inherently difficult because fertilizers are:

(a) Uniform and dry
(b) Hygroscopic, corrosive, variable in flowability
(c) Dense only
(d) Fibrous

Q127. The feed used extensively on grain-drill fertilizing units is:

(a) Vacuum disc
(b) Star-wheel feed
(c) Cell plate
(d) Spinning disc

Q128. In star-wheel feed, application rate is adjusted by:

(a) Fan speed
(b) Gate opening and wheel size/speed
(c) Boom height
(d) Nozzle size

Q129. Auger (screw) feeder delivery rate is proportional to:

(a) Screw speed and pitch volume
(b) Hopper height
(c) Gate colour
(d) Tube length

Q130. The single-pass distribution pattern of a spinning-disc broadcaster is:

(a) Rectangular
(b) Triangular/oval, requiring 25–35% overlap
(c) Ring
(d) Square

Q131. In spinner spreading, fine particles:

(a) Travel farthest
(b) Fall short — causing pattern segregation
(c) Do not move
(d) Bounce back

Q132. Particle velocity leaving a spinner disc tip equals approximately:

(a) $\omega \cdot r$
(b) ω / r
(c) r / ω
(d) $\omega^2 r$

Q133. Recommended fertilizer band position relative to seed:

(a) In contact
(b) 5 cm to the side and 5 cm below
(c) 15 cm above
(d) On the surface only

Q134. Direct seed–fertilizer contact is avoided because:

(a) Blocking tubes
(b) Salt injury to germinating seed
(c) Cost
(d) Corrosion of opener

Answers

Q126 — (b) Hygroscopic, corrosive, variable in flowability
Q127 — (b) Star-wheel feed
Q128 — (b) Gate opening and wheel size/speed
Q129 — (a) Screw speed and pitch volume
Q130 — (b) Triangular/oval, requiring 25–35% overlap
Q131 — (b) Fall short — causing pattern segregation
Q132 — (a) $\omega \cdot r$
Q133 — (b) 5 cm to the side and 5 cm below
Q134 — (b) Salt injury to germinating seed

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 11 (Crop Planting), Ch. 12 (Row-Crop Cultivation), Ch. 13 (Application of Fertilizers); standard FMP design references.